

# Rama Krishna Aditya Bharadwaj Kolluri, MSc

Doctoral Researcher | Business Intelligence Manager | Mechatronics Engineer

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## Profile

Mechatronics engineer and Business Intelligence Manager at ECAM Engineering Limited, currently undertaking a PhD at Birmingham City University on AI-driven maintenance optimisation for solar minigrids (UKRI EPSRC Ayrton Challenge). I deliver data-driven digital transformation across manufacturing operations, from shop-floor data capture to executive planning. My focus is operational efficiency, market intelligence, and the practical application of analytics and AI to support sound business decisions.

## Technical Skills

- **Machine Learning / AI:** TensorFlow, OpenCV, neural networks, time-series forecasting (ARIMA)
- **Data Analysis:** Python (Pandas, NumPy, Matplotlib, scikit-learn), SQL
- **Software Development:** Python, PHP, Java; MySQL; Git
- **Testing & Automation:** Selenium, Postman, Python unit testing
- **Simulation & Hardware:** MATLAB, NI LabVIEW, Arduino, Raspberry Pi, ESP32, sensor integration
- **Domains:** Industry 5.0, cyber-physical systems, IoT, MES, OEE, MRP-II
- **Project Management:** Agile

## Certifications

Agile Project Management (Foundation & Practitioner) · Business Agility Professional Level 1 · BCS Business Analysis: Essential Tools and Techniques · Neural Networks and Deep Learning · ChatGPT Prompt Engineering for Developers · Effective Communications

## Experience

### Doctoral Researcher — Birmingham City University, Birmingham

February 2026 – February 2029

- Researching AI-driven predictive maintenance optimisation for solar minigrids under the UKRI EPSRC Ayrton Challenge programme.
- Focus areas: predictive maintenance, digital twins, and energy access in off-grid contexts.

### Business Intelligence Manager — ECAM Engineering Ltd, Cheadle, Stoke-on-Trent

August 2025 – Present

- Lead ECAM's data strategy from shop-floor operations to strategic planning.
- Maintain production data; track capacity, OEE, and KPIs to surface improvement opportunities.
- Build dashboards and reports that inform decision-making across the business.
- Developed and maintain a lightweight MRP-II system giving extended visibility into production schedules and forecasts for multi-level bills of materials.
- Monitor industry and competitor trends; advise leadership on market gaps and AI-led efficiency gains.

### Business Systems and Technology Analyst (KTP Associate) — Birmingham City University & ECAM Engineering Ltd

July 2023 – July 2025

- Led a digitalisation project improving visibility and traceability across the manufacturing process.
- Embedded academic knowledge from Birmingham City University into ECAM's operations, bridging university research and industry practice.
- Analysed manufacturing systems to develop technical solutions aligned to both research and business objectives.
- Transferred real-world manufacturing challenges back to the university, contributing to the academic knowledge base.

### Graduate Project Engineer (DigitME2) — University of Central Lancashire, Burnley

February 2022 – June 2023

- Developed features for the DigitME2 web application (PHP, MySQL) and Android application (Java).
- Built a Process Time Tracker (MES), inventory tracker, and OEE monitoring tools for SMEs.
- Automated testing with Python unit tests and Selenium.
- Built machine-learning models (ARIMA) for time-series prediction using simulated datasets.

### **Research Intern — Autonomous Vehicles (Erasmus+ Traineeship) — Arab Academy for Science, Technology & Maritime Transport, Alexandria, Egypt**

June 2021 – September 2021 · Grade: Distinction · European Commission funded

- Researched low-cost object detection (ultrasonic, infrared, camera modules) for autonomous navigation.
- Integrated Arduino UNO and Raspberry Pi with MPU-6050 sensors for lane-following and vehicle stability.
- Implemented MATLAB and Python algorithms for pothole detection and obstacle avoidance.

### **Earlier Roles**

- Assembly Operator, Baxi Heating, Preston (Nov 2021 – Feb 2022)
- Production & Warehouse Operator, Royal Sanders (UK) Ltd, Preston (May 2020 – Sep 2020)
- Sortation Associate, Amazon Logistics, Preston (Sep 2020 – Jan 2021)
- Customer Service Associate, Amazon Development Center India, Hyderabad (Aug 2018 – Aug 2019)

### **Education**

**PhD, AI-Driven Maintenance Optimisation for Solar Minigrids** — Birmingham City University · 2026 – 2029

**MSc, Mechatronics and Intelligent Machines (Distinction)** — University of Central Lancashire, Preston · 2020 – 2021

- Modules: IoT, AI, sensors, control systems; tools: MATLAB, NI LabVIEW.
- Project: Smart autonomous planetary rover with drive-chain alignment.
- Group project: Piezoelectric energy harvester for low-cost remote sensors.

**BTech, Mechanical Engineering** — Vishnu Institute of Technology, Bhimavaram, India · 2013 – 2018

- Project: Low-cost box transfer machine for small-scale industries.

### **Achievements**

- First place, Robo Car Race, Vishnu Institute of Technology, November 2016.

### **Languages**

English (Full Professional) · Telugu (Native) · Hindi (Limited Working)